DATA SCIENCE 1 · 2026–2027

Tell the story with plotly

Syllabus for lab 5 – *Data Visualization I*

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Session	lab 5 – *Data Visualization I*

1 Overview

In labs 3, 4 and 5 you are the new data analyst of a (fictional) youth football and athletics club. The head coach gives you three export files from the club's systems. They are messy on purpose, the way real exports are: inconsistent labels, sprint times in milliseconds, missing values, an athlete who appears twice and a column with a comma as the decimal separator.

Each session builds on the previous one:

Lab 3 **From file to tidy table.** Load, explore, filter, add columns and clean until you can trust the data.

Lab 4 **Combine, summarize, reshape.** Summarize groups, combine tables, switch between a wide and a long shape, and work with a heart-rate series per second.

Lab 5 **Tell the story with plotly.** Choose the right chart for a question, make honest charts with `plotly.express` and combine them into a small interactive dashboard.

2 Learning outcomes

After lab 5 you can:

LO1 choose a chart type that fits the question, and recognise misleading charts, such as a truncated axis;

LO2 make histograms, box plots, scatter plots and line charts with `plotly.express`, with colour, facets and size for extra dimensions;

LO3 make charts interactive with marimo UI elements and combine them into a dashboard.

3 Prerequisites

Labs 3 and 4 (*Data Processing I and II*): loading, filtering and cleaning data with `pandas`, grouping with `groupby`, combining tables with `merge`, and the difference between a long and a wide shape. Lab 5 starts with the complete cleaning of labs 3 and 4 in one cell.

4 Software and materials

Every lab is a **marimo notebook** in the `labs/` folder of the course repository. You can open it in two ways:

- **Online:** go to `https://marimo.app`, choose *New* → *Open from URL...* and paste the link to the notebook you get from your lecturer. The data is then read online, so you don't have to install anything.

- **Locally:** install `uv` and Python 3.12 following the *Installing Python* manual (or the slides). Then go to the course folder in a terminal and open a lab with:
  `uv run marimo edit labs/lab3_data_processing_I.py`
  The first time, uv installs the packages it needs; the data is read from the `data/` folder.

Three things to know about marimo:

- Notebooks are **reactive**: when you change a cell, every cell that depends on it runs again, in the order of their dependencies.
- **Two cells cannot define the same variable.** Give every step its own clear name, such as `tests_raw` and `tests_clean`, or keep a sequence of steps together in one cell.
- Every exercise contains a `...` placeholder. A hidden cell below it checks your answer: it turns **green** when the answer is right, and otherwise gives a general hint about what is still wrong, without giving the answer away.

5 The dataset

File	Size	Content and known problems
`athletes.csv`	43 rows, 7 columns	ID, name, birthdate, gender, position, height, weight. 13 spellings for 4 positions, weight as text with a decimal comma (`"57,5"`), one athlete entered twice with two IDs, a few missing values.
`tests.csv`	114 rows, 8 columns	10 m and 30 m sprint, countermovement jump height and Yo-Yo level, at 3 test moments. Some sprint times are stored in milliseconds; a few missing values.
`hr_session.csv`	12,000 rows, 5 columns	Heart rate and speed per second for 5 athletes during one 40-minute training session.

6 Schedule

Goal: choose a chart type for a given question, make the standard `plotly.express` charts, add extra dimensions with colour, facets and size, and build in simple interactivity.

Time	Block	What you practise
13:00–13:30	Principles	The right chart for the question (distribution, comparison, relationship, evolution); critique a misleading chart.
13:30–14:30	`plotly.express` basics	Histogram, box plot, scatter plot and line chart, with titles, axis labels and hover information.
14:30–15:15	More dimensions	`color`, `facet_col` and `size`; one fixed colour per position.
15:15–15:30	Break	
15:30–16:15	Heart rate as a time series	Zone bands with `add_hrect`, and why we don't use a second y-axis.
16:15–16:45	Interactivity	`mo.ui.dropdown`, `mo.ui.range_slider`, optionally selections with `mo.ui.plotly`.
16:45–17:30	Mini-dashboard	3 charts and 1 selector that at least one chart responds to.

Exercises: (1) a histogram of weight; (2) Yo-Yo level per position, with facets per gender; (3) speed over time for one athlete; **assignment:** the mini-dashboard.

7 Working method

- Each block starts with a short demo in the notebook, followed by exercises you solve in the notebook itself.
- Use the automatic checks as feedback, not as the goal: when a check turns green, make sure you can explain *why* your code works.
- You build the mini-dashboard with everything from this lab: the goal is combining, not learning new chart types.

8 Assessment

- **Mini-dashboard:** three charts and one interactive selector, made with the club's data. It can serve as your submission for this module and prepares the case presentations and the Project Lab next semester.

9 Preparation

- Check before the session that you can open the notebook, locally or on marimo.app.
- Review `groupby`, `merge` and the long shape from lab 4. Charts with plotly express start from data in the long shape.
- Read the course text for lab 5 next to the notebook.